

Introduction

- In this section we will develop a system of rule for manipulating formulas in symbolic logic.
- This system, called the propositional calculus, will allow us to make logical deductions formally.

Introduction

Definition:

- A statements that are **always true**, no matter what the T/F values of the component statements are, **is called a tautology**, and we write

$$(p \wedge q) \Rightarrow p$$

- The notation $A \Rightarrow B$ means that the statement $A \rightarrow B$ is true in all cases; in other words, the truth table for $A \rightarrow B$ is all T's. Similarly, the $\Leftrightarrow$ symbol denotes a tautology containing the $\leftrightarrow$ connective.
- There are also statements in formal logic that **are never true**. A statement whose truth table contains all F's **is called a contradiction**.
- A statement in propositional logic that is **neither a tautology nor a contradiction** **is called a contingency**. A contingency has both T's and F's in its truth table.

Tautology

Example 1:

Use a truth table to show $(p \wedge q) \rightarrow p$ is a tautology

Tautology

Solution

p	q	$p \wedge q$	$(p \wedge q) \rightarrow p$
T	T	T	T
T	F	F	T
F	T	F	T
F	F	F	T

Contradiction

Example 2:

Use a truth table to show $p \wedge \neg p$ is a contradiction.

Contradiction

Solution

p	$\neg p$	$p \wedge \neg p$
T	F	F
F	T	F

Contingency

Example 3:

Use a truth table to show $p \wedge q$ is a contingent

Contingency

Solution

p	q	$p \wedge q$
T	T	T
T	F	F
F	T	F
F	F	F

Equivalences Rules

- Tautologies are important because they show how one statement may be logically deduced from another.
- For example, suppose we know that the following statements are true.

Our professor does not own a spaceship.

If our professor is from Mars, then our professor owns a spaceship.

- Every tautology can be used as a rule to justify deriving a new statement from an old one.
- There are two types of derivation rules:
 - equivalence rules.
 - inference rules.
- Equivalence rule describe logical equivalences.
- Inference rules describe when a weaker statement can be deduced from a stronger statement.

Equivalences Rules

- The equivalence rules given in following Table could all be checked using truth tables.
- If A and B are statements (possibly composed of many other statements joined by connectives), then the tautology $A \Leftrightarrow B$ is another way of saying that A and B are logically equivalent.

Equivalence	Name
$p \Leftrightarrow \neg \neg p$	double negation
$p \rightarrow q \Leftrightarrow \neg p \vee q$	implication
$\neg(p \wedge q) \Leftrightarrow \neg p \vee \neg q$ $\neg(p \vee q) \Leftrightarrow \neg p \wedge \neg q$	De Morgan's laws
$p \vee q \Leftrightarrow q \vee p$ $p \wedge q \Leftrightarrow q \wedge p$	commutativity
$p \wedge (q \wedge r) \Leftrightarrow (p \wedge q) \wedge r$ $p \vee (q \vee r) \Leftrightarrow (p \vee q) \vee r$	associativity

Inference Rules

Modus ponens

- When a tautology is of the form $(C \wedge D) \Rightarrow E$, we often prefer to write

$$\left. \begin{array}{l} C \\ D \end{array} \right\} \Rightarrow E$$

This notation highlights the fact that if you know both C and D , then you can conclude E .

Example 4: use a truth table to prove the following.

$$\left. \begin{array}{l} p \\ p \rightarrow q \end{array} \right\} \Rightarrow q$$

Solution:

let S be the statement $[p \wedge (p \rightarrow q)] \rightarrow q$. We construct our truth table by building the parts of S as following

Inference Rules

p	q	$p \rightarrow q$	$p \wedge (p \rightarrow q)$	S
T	T	T	T	T
T	F	F	F	T
F	T	T	F	T
F	F	T	F	T

- Since the column for S is all T's, this proves that S is a tautology. the tautology in [Example 4](#) is known as modus ponens. We can stated it as follows.

If the **first**, then the **second**;
but the **first**;

Example: Therefore the **second**.

- "If you have a current password, then you **can log on to the network**"
- "You have a current password"

Therefore:

- "You can log on to the network"

Inference Rules

Modus tollens

- When a tautology is of the form

$$\left. \begin{array}{l} \neg q \\ p \rightarrow q \end{array} \right\} \Rightarrow \neg p$$

- Inference rules work in only one direction. An inference rule of the form $A \Rightarrow B$ allows you to do only one thing:

1. Given A , deduce B .

Example:

- You can't log into the network
- If you have a current password, then you can log into the network

Therefore

- You don't have a current password.

Inference Rules

Inference	Name
$\frac{p \quad q}{p \wedge q}$	conjunction
$\frac{p \quad p \rightarrow q}{q}$	<i>modus ponens</i>
$\frac{\neg q \quad p \rightarrow q}{\neg p}$	<i>modus tollens</i>
$p \wedge q \Rightarrow p$	simplification
$p \Rightarrow q \vee q$	addition

Example: write a proof sequence for the assertion

$$\frac{p \quad p \rightarrow q \quad q \rightarrow r}{r}$$

Inference Rules

$$\left. \begin{array}{l} p \\ p \rightarrow q \\ q \rightarrow r \end{array} \right\} \Rightarrow r.$$

Solution:

Statements	Reasons
1. p	given
2. $p \rightarrow q$	given
3. $q \rightarrow r$	given
4. q	<i>modus ponens</i> , 1, 2
5. r	<i>modus ponens</i> , 4, 3

Inference Rules

Example:

Use truth tables to establish the *modus tollens* tautology:

$$\left. \begin{array}{l} \neg q \\ p \rightarrow q \end{array} \right\} \Rightarrow \neg p$$

Solution:

p	q	$\neg q$	$p \rightarrow q$	$(\neg q) \wedge (p \rightarrow q)$	$\neg p$	$[(\neg q) \wedge (p \rightarrow q)] \rightarrow \neg p$
T	T	F	T	F	F	T
T	F	T	F	F	F	T
F	T	F	T	F	T	T
F	F	T	T	T	T	T

Exercises

Show that the following are tautologies:

1. $\neg(p \vee q) \Leftrightarrow \neg p \wedge \neg q$

2. $a \Rightarrow (b \Rightarrow a)$

Show that $p \Leftrightarrow (p \Rightarrow \neg p)$ is a contradiction:

Show that $p \vee q$, and $p \rightarrow q$ are all contingencies

Show that $p \Leftrightarrow \neg\neg p$